Inventory Management System and Stock Lifecycle

---

1. Preface

This document specifies the **Software Requirements Specification (SRS)** for the **Inventory Management System (IMS)**.
It is intended to provide a comprehensive description of system functionality, constraints, and behavior for all stakeholders including system users, developers, testers, and evaluators to understand what the system should do.

The document serves as a baseline agreement between stakeholders and the development team regarding system expectations and deliverables.

---

2. Introduction

2.1 Purpose

The purpose of this document is to define the **functional and non-functional requirements** of the Inventory Management System based on stakeholder needs and system modeling using **use-case diagrams, activity diagrams, and sequence diagrams**.

This SRS ensures clarity, consistency, and traceability across system design, implementation, and validation phases.

2.2 Scope

The Inventory Management System (IMS) is a **centralized, analytics-driven platform** designed to support retailers in managing:

- Product catalogs
- Inventory levels
- Supplier coordination
- Consumer availability
- Multi-store inventory operations

The system integrates multiple stakeholders and external POS systems to ensure **real-time stock visibility**, **optimized replenishment**, and **efficient supply chain operations**.

IMS is not responsible for low-level POS hardware control but relies on standardized interfaces for data synchronization.

2.3 Definitions, Acronyms, and Abbreviations

Term	Description
IMS	Inventory Management System
Retailer	Business entity managing inventory and sales
Supplier	Entity responsible for product supply and fulfillment
Consumer	End user checking availability and providing feedback
POS	Point of Sale system
Admin	System administrator with governance privileges
Stockout	Situation where inventory reaches zero
Reorder Point	Threshold triggering replenishment

---

3. Overall Description

3.1 System Users

The system supports the following user roles:

- Retailer
- Supplier
- Consumer
- System Administrator
- POS System (External)

3.2 Product Perspective

IMS is a **web-based centralized system** that integrates:

- Inventory tracking
- Supplier management
- Consumer interaction
- POS data synchronization
- Analytics and reporting

The system follows a modular architecture with clearly defined role-based access control and supports horizontal scaling for multi-store deployments.

3.3 Design and Implementation Constraints

- POS data depends on third-party API availability
- Real-time updates require stable network connectivity
- Data consistency must be preserved across multiple store locations

---

4. User Requirements (High-Level)

ID	User Requirement
UR1	The system shall allow retailers to manage products and inventory
UR2	The system shall provide real-time stock visibility
UR3	The system shall generate intelligent reorder recommendations
UR4	The system shall support multi-store inventory management
UR5	The system shall integrate with POS systems

UR6	The system shall allow suppliers to manage order fulfillment and delivery
UR7	The system shall notify consumers of stock availability
UR8	The system shall provide analytics and audit reports

---

5. Functional Requirements

5.1 Retailer Functions

- **FR1:** The system shall allow retailers to add, update, and remove product details.
- **FR2:** The system shall maintain real-time inventory levels per product and store.
- **FR3:** The system shall track inventory movement from procurement to sale and return.
- **FR4:** The system shall recommend reorder quantities based on demand trends.
- **FR5:** The system shall forecast future demand using historical sales data.
- **FR6:** The system shall support centralized inventory management across multiple stores.

5.2 Supplier Functions

- **FR7:** The system shall allow suppliers to confirm order fulfillment.
- **FR8:** The system shall track supplier delivery status in real time.
- **FR9:** The system shall evaluate supplier performance based on delivery metrics.
- **FR10:** The system shall allow suppliers to manage delivery schedules and capacity.

5.3 Consumer Functions

- **FR11:** The system shall allow consumers to view real-time product availability.

- **FR12:** The system shall notify consumers when out-of-stock products are restocked.
- **FR13:** The system shall allow consumers to submit product feedback.

5.4 Administrator Functions

- **FR14:** The system shall allow administrators to configure multi-store structures.
- **FR15:** The system shall manage user roles and access permissions.
- **FR16:** The system shall control global system configurations.
- **FR17:** The system shall generate system-wide analytics and audit reports.

5.5 POS Integration Functions

- **FR18:** The system shall automatically update inventory based on POS sales data.
- **FR19:** The system shall ensure synchronization consistency between POS and inventory records.

---

6. Non-Functional Requirements

6.1 Performance Requirements

- **NFR1:** The system shall respond to user actions within 3 seconds.
- **NFR2:** Inventory updates from POS shall reflect within 2 seconds.

6.2 Security Requirements

- **NFR3:** The system shall enforce role-based access control.
- **NFR4:** Only authenticated users shall access system functions.

6.3 Reliability Requirements

- **NFR5:** The system shall ensure data consistency across stores.
- **NFR6:** The system shall recover gracefully from system failures.

6.4 Usability Requirements

- **NFR7:** The system shall provide a simple and intuitive interface.
- **NFR8:** Minimal training shall be required for system usage.

6.5 Scalability Requirements

- **NFR9:** The system shall support onboarding of additional stores without downtime.

---

7. System Models

The system is modeled using:

- Use Case Diagrams – Actor interactions
- Activity Diagrams – Workflow logic
- Sequence Diagrams – Inventory lifecycle interactions

---

8. Appendices

8.1 Hardware Requirements

- Server: Minimum 8-core CPU, 32GB RAM, 1TB SSD
- Client: Standard web-enabled devices

8.2 Database Requirements

- Persistent storage of inventory, sales, and audit logs
- Historical data retention for analytics and forecasting

---

9. Conclusion

This SRS defines the complete functional scope, constraints, and future direction of the Inventory Management System, ensuring alignment between stakeholders and providing a solid foundation for design, development, and validation.

10. Index

To ensure scannability for different stakeholders, the following indexes are included:

• Index of Diagrams:

○ **Use Case Diagram** (Page 9)

○ **Activity Diagrams** (Page 10–15)

– Managing Inventory

– Analyze Inventory & Audit Data

– Monitor Supplier Performance

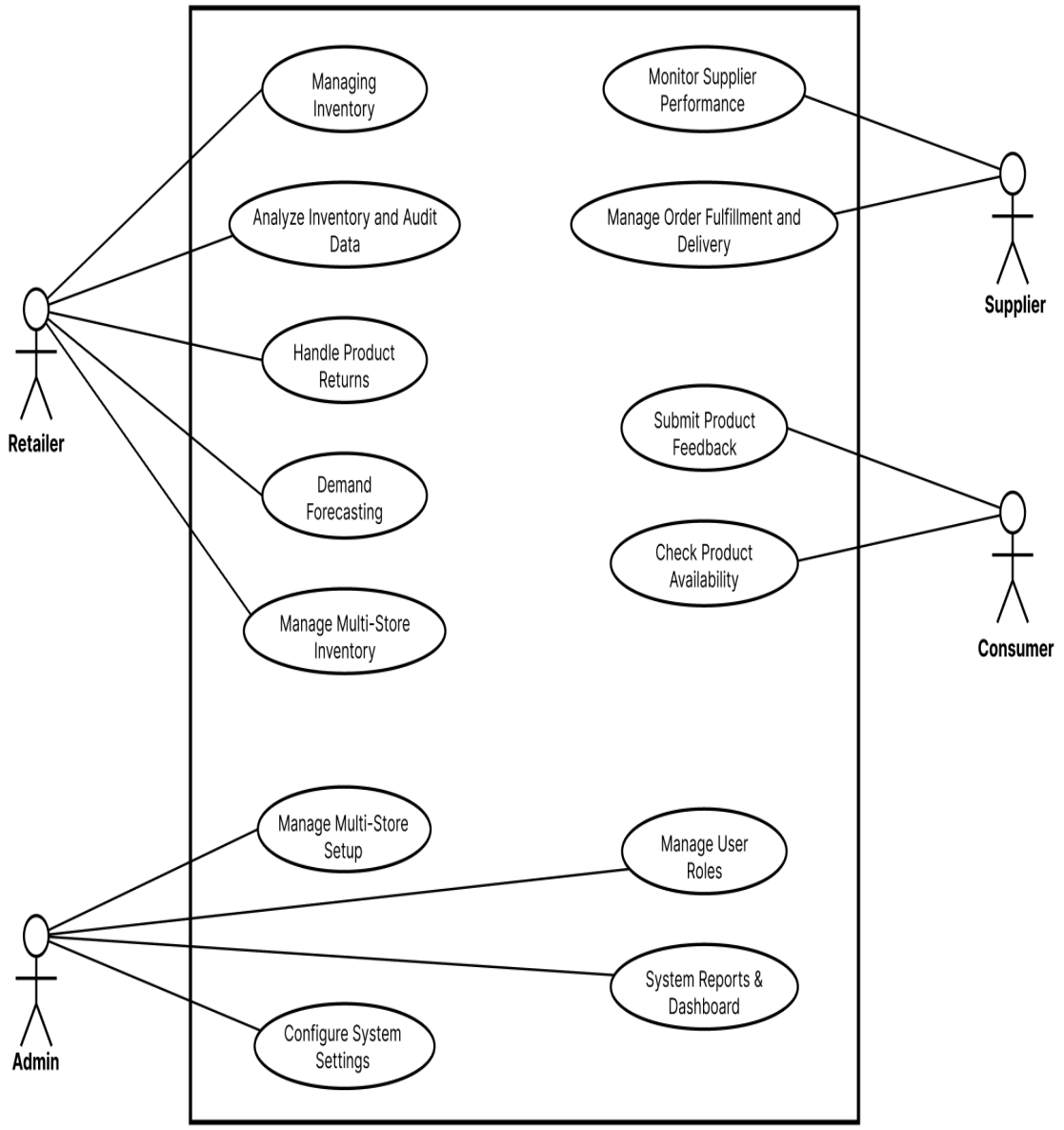
– Product Availability

– Product Feedback

– Manage Order Fulfillment and Delivery

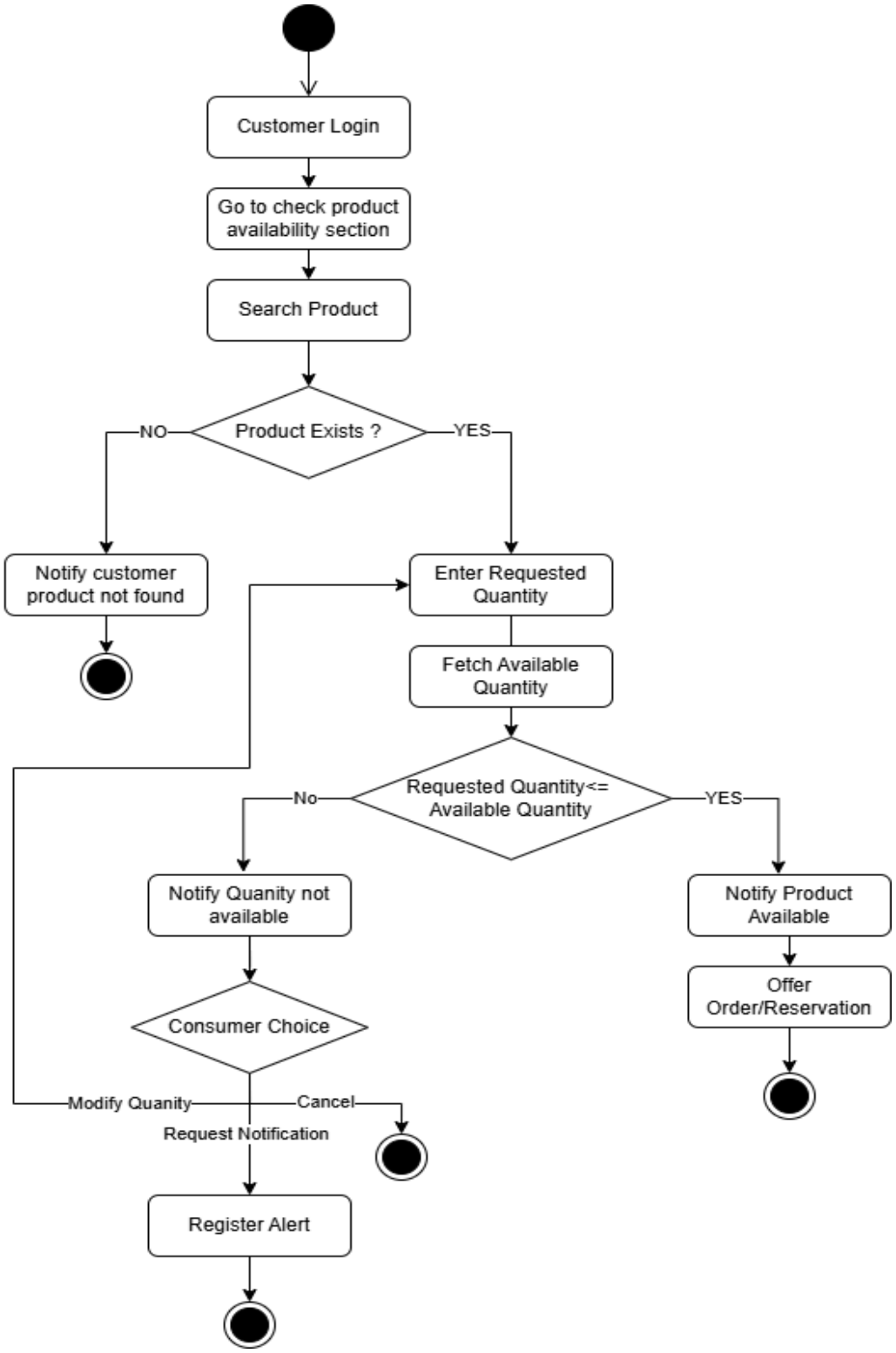
- **Sequence Diagram** (Page 16)

Use case Diagram:

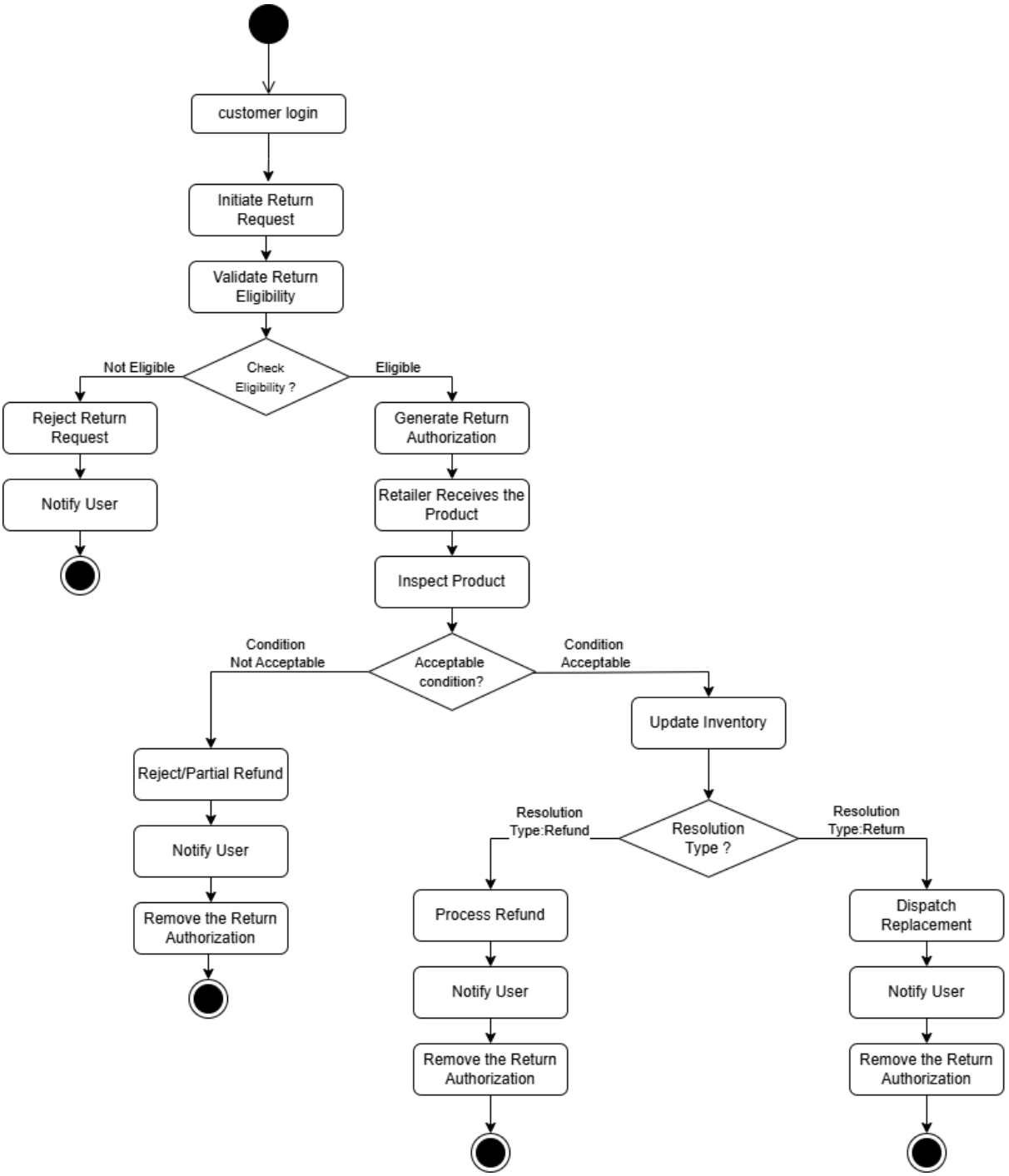


Activity Diagrams:

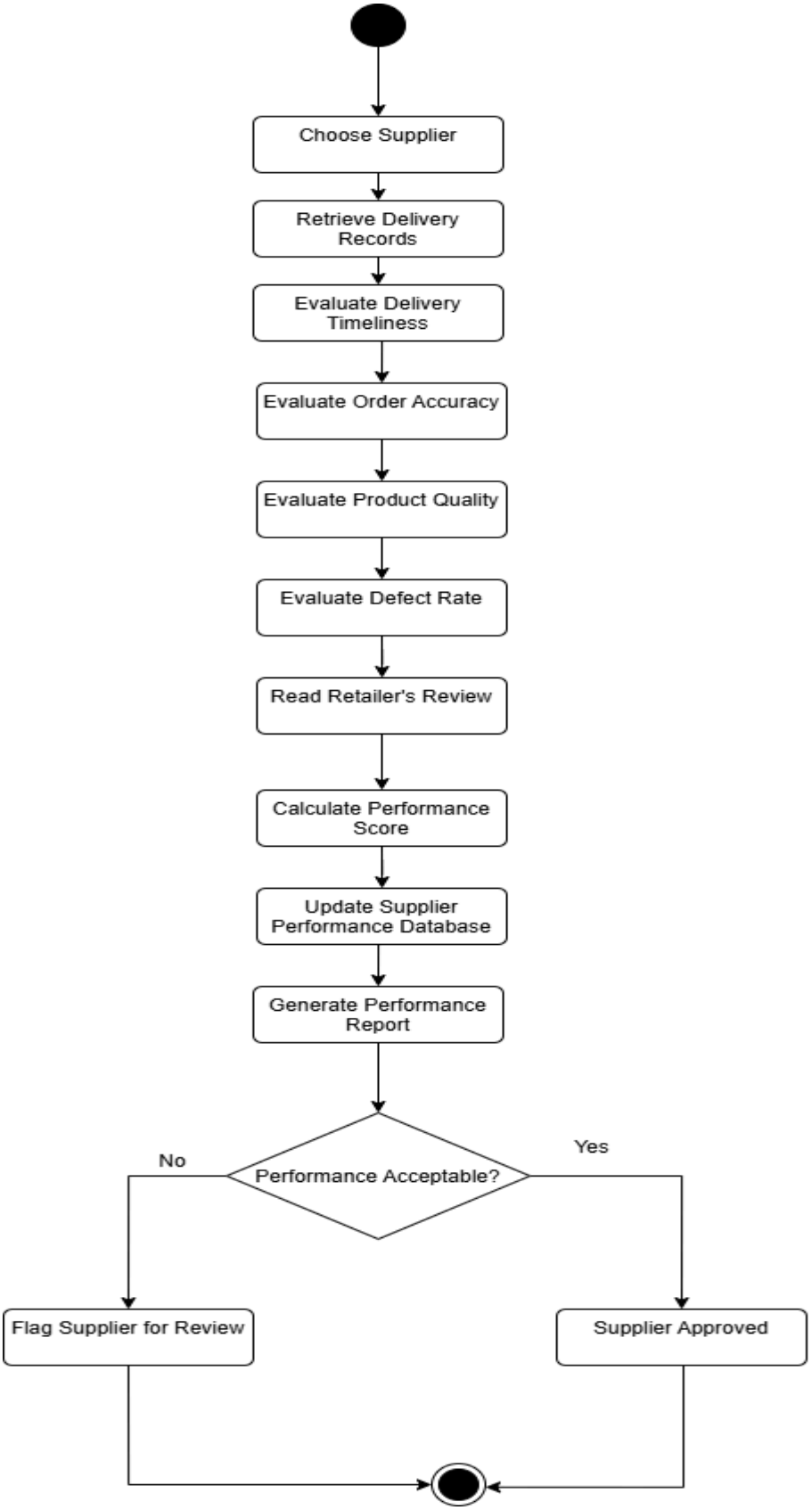
1.) Product Availability:



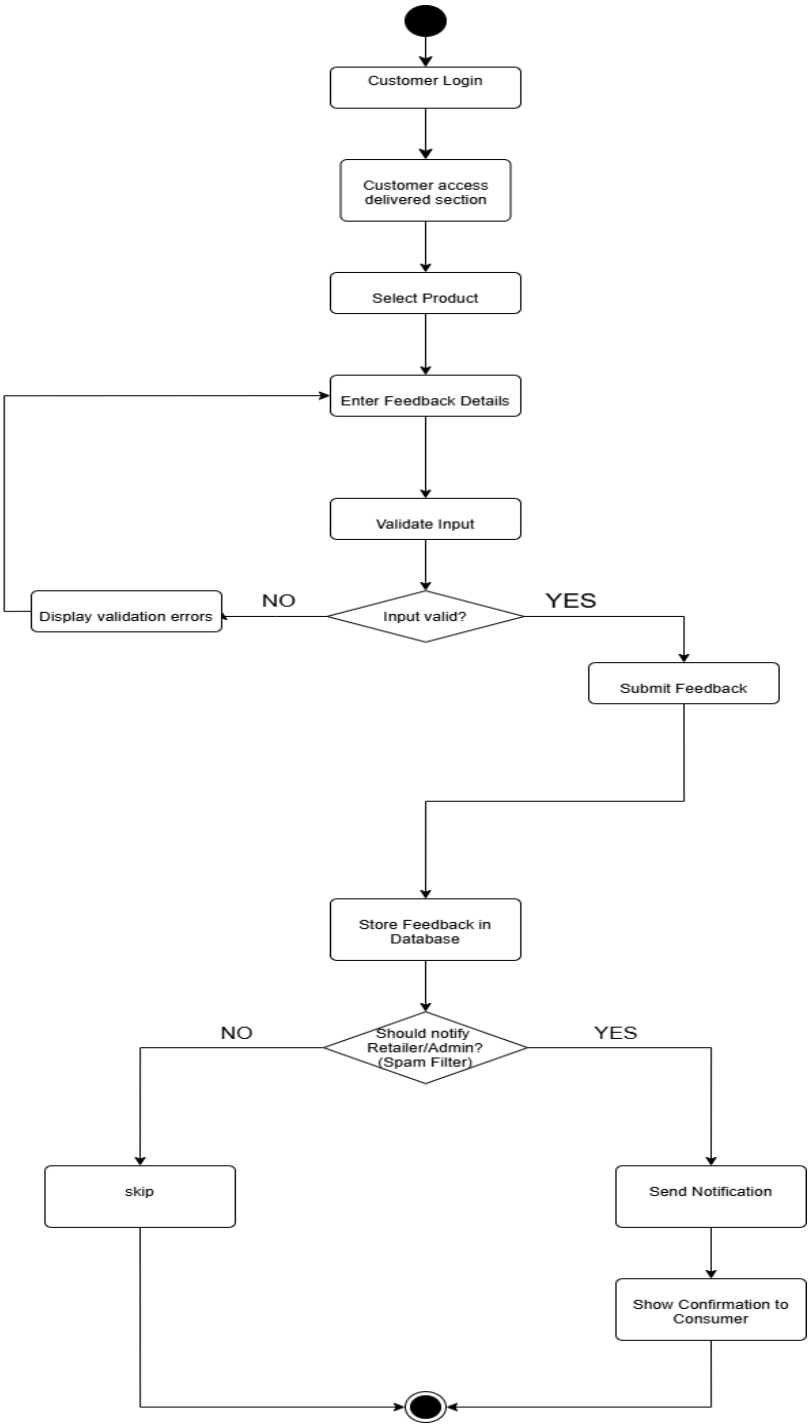
2.) Returns Management:



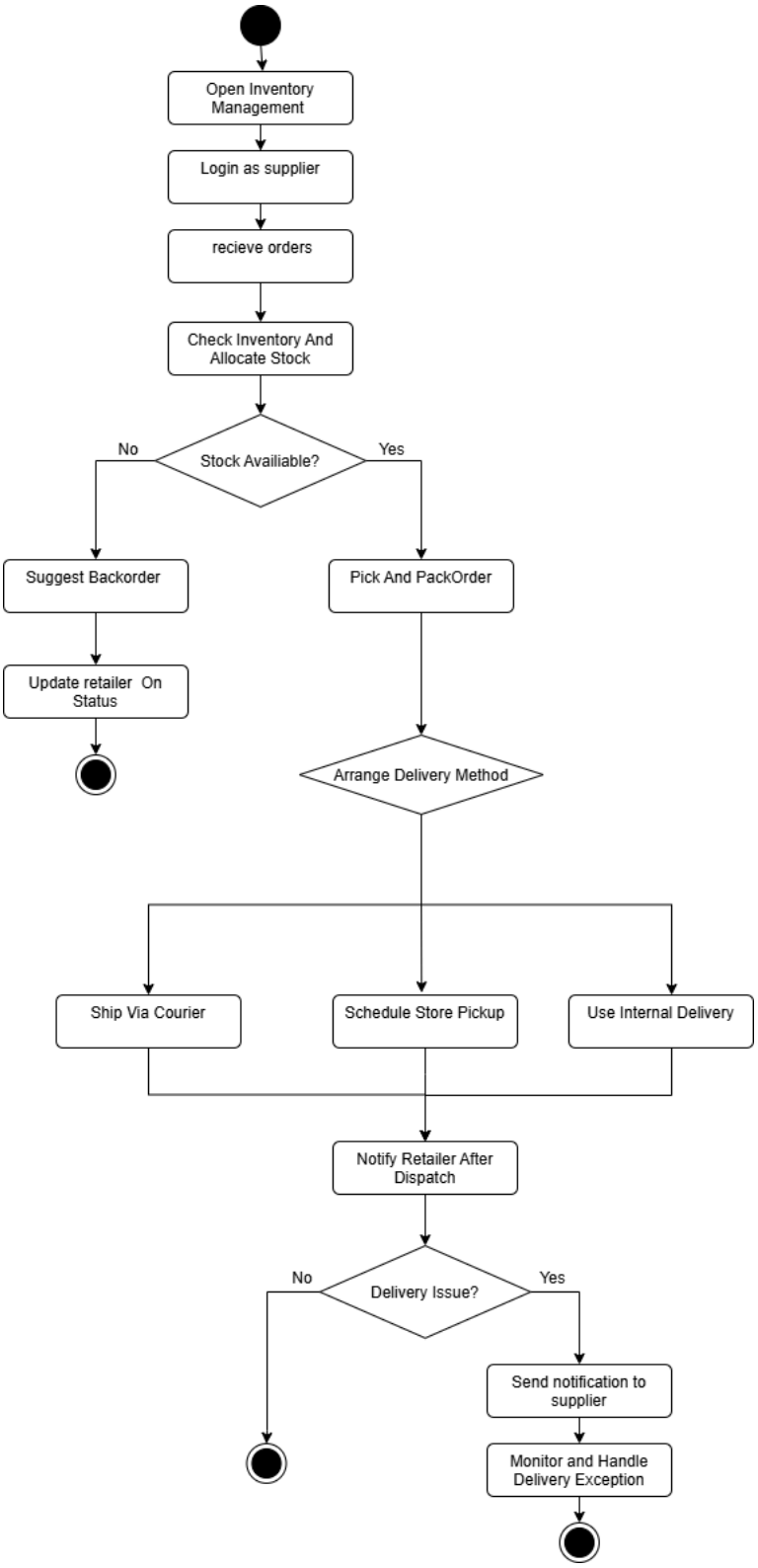
3.) Supplier Performance:



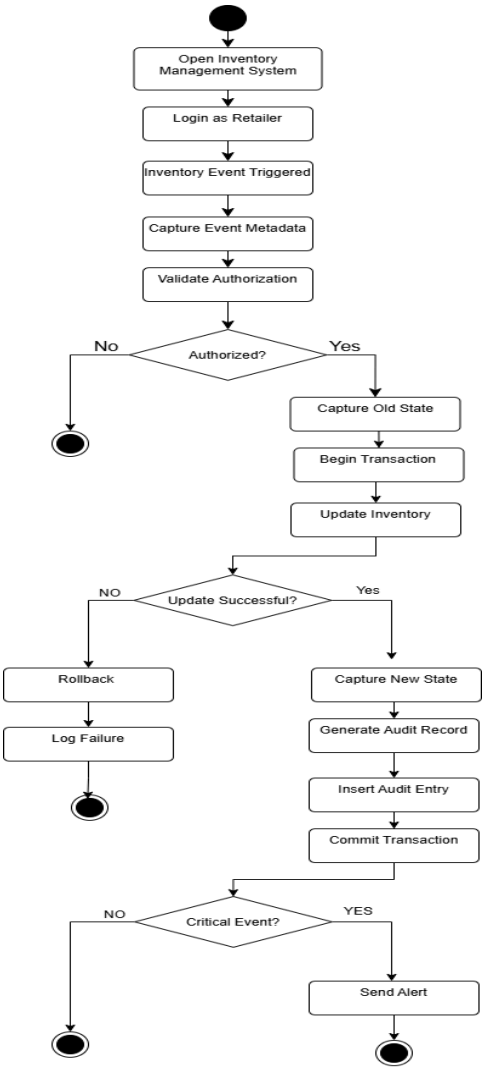
4.)product feedback:



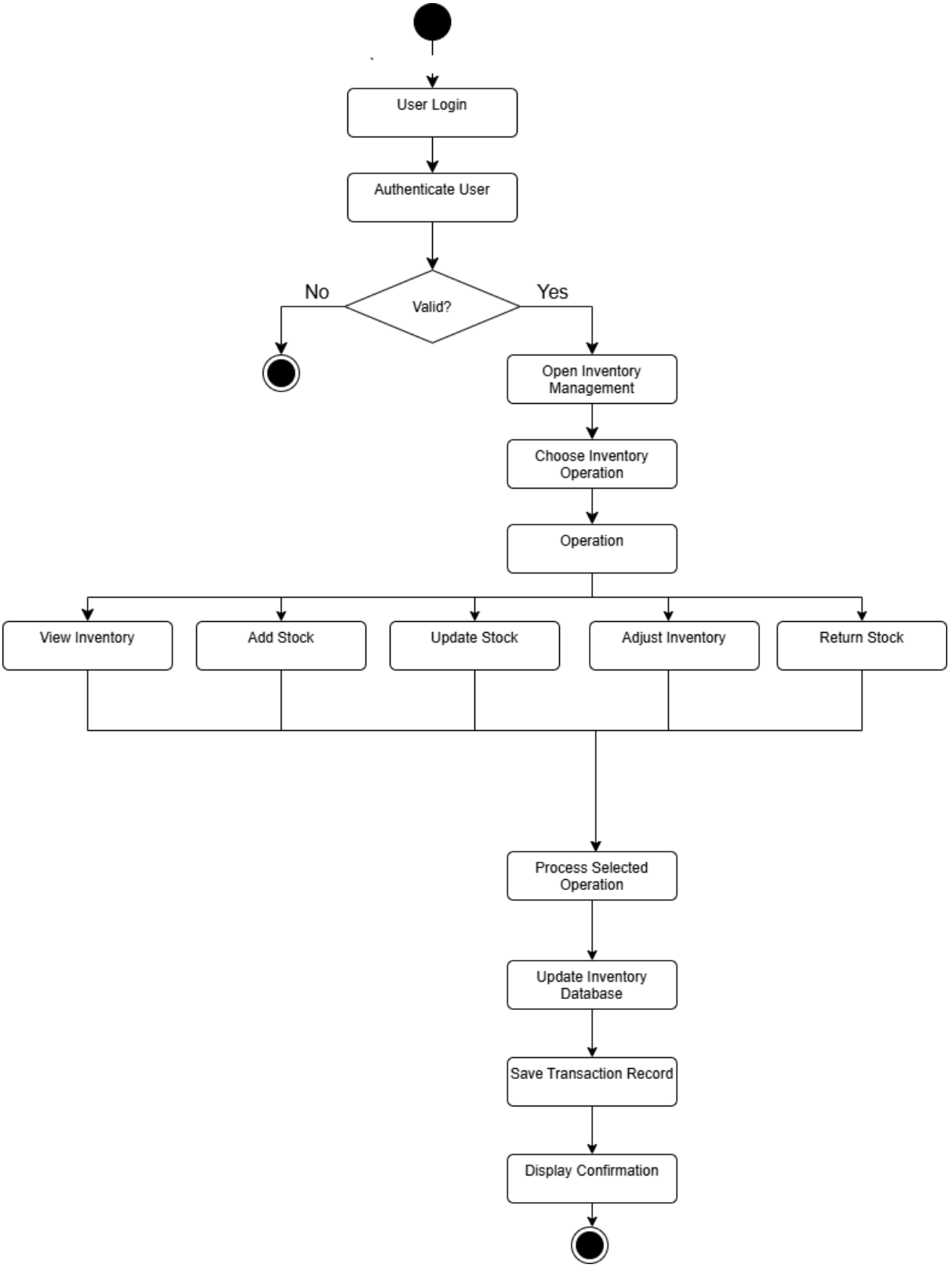
5.) order fulfillment:



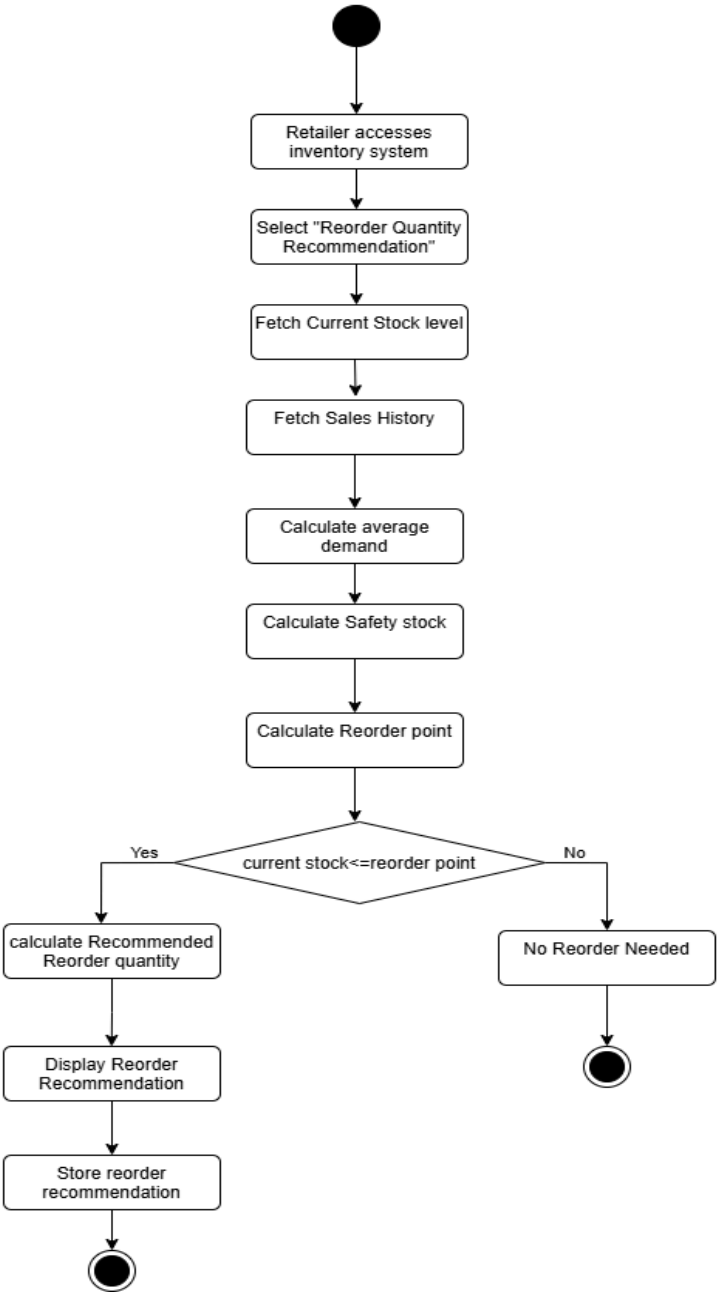
6.) Audit data:


Open Inventory Management System
Login as Retailer
Inventory Event Triggered
Capture Event Metadata
Validate Authorization
Authorized?
No
Yes
Capture Old State
Begin Transaction
Update Inventory
Update Successful?
NO
Yes
Rollback
Log Failure
Capture New State
Generate Audit Record
Insert Audit Entry
Commit Transaction
Critical Event?
NO
YES
Send Alert

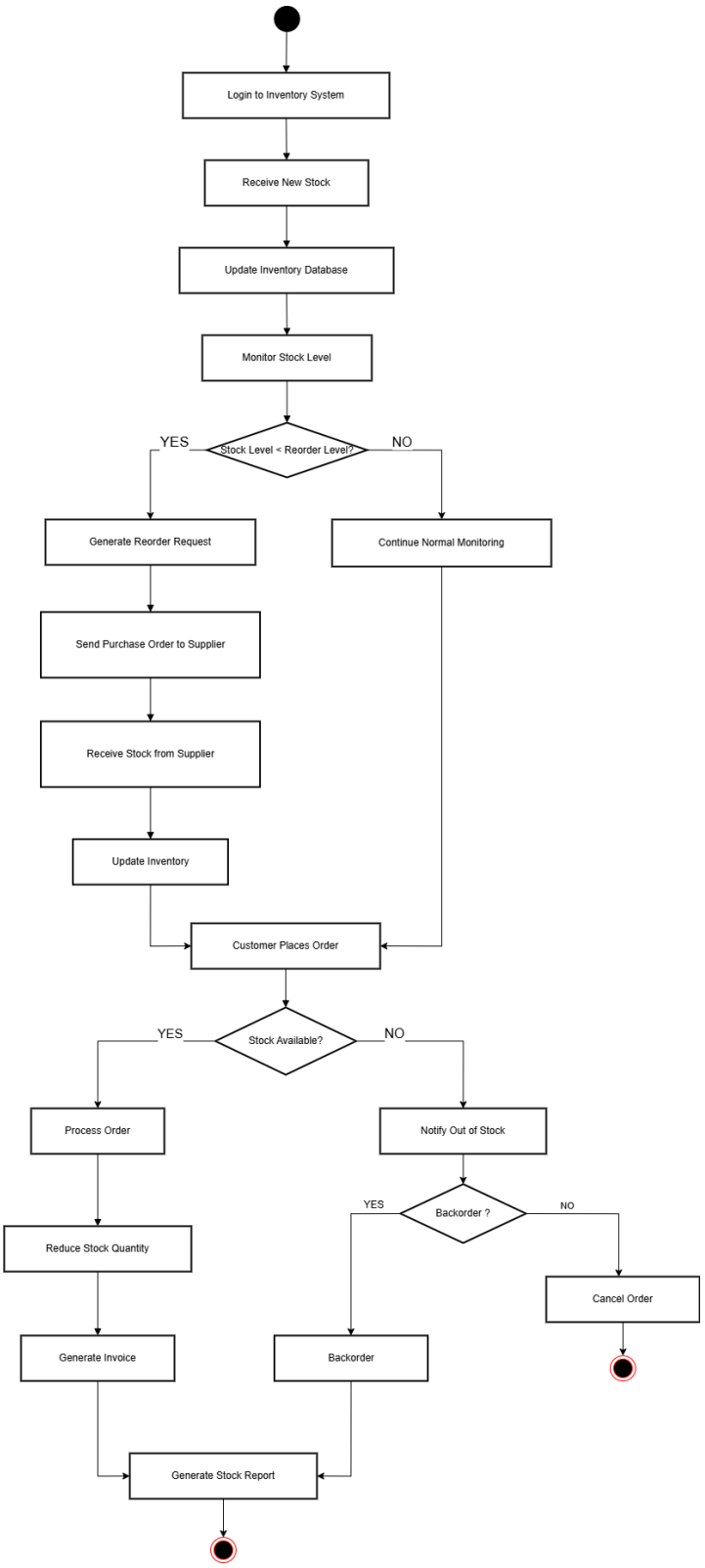
7.) manage inventory:



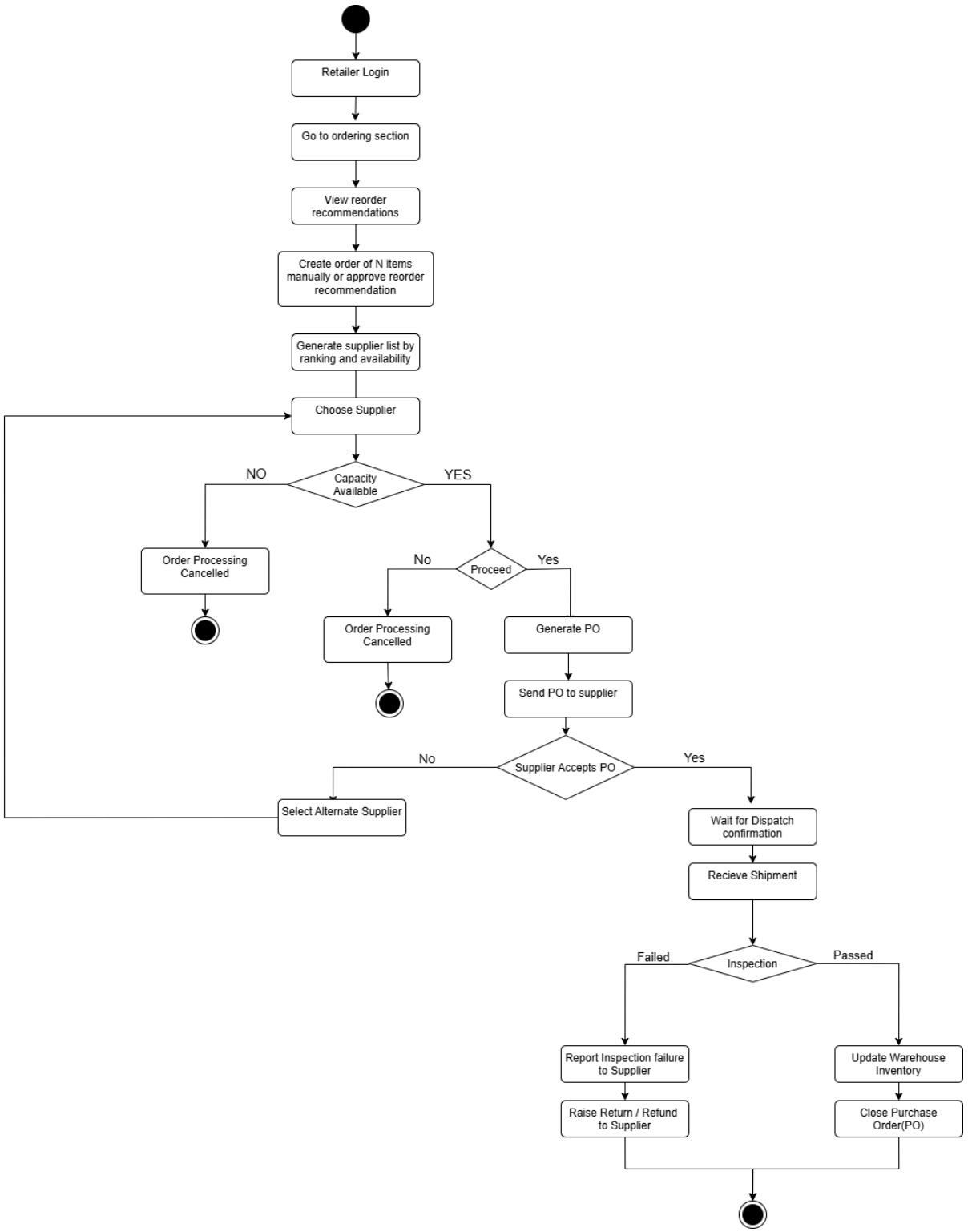
8.) Reorder Recommendation



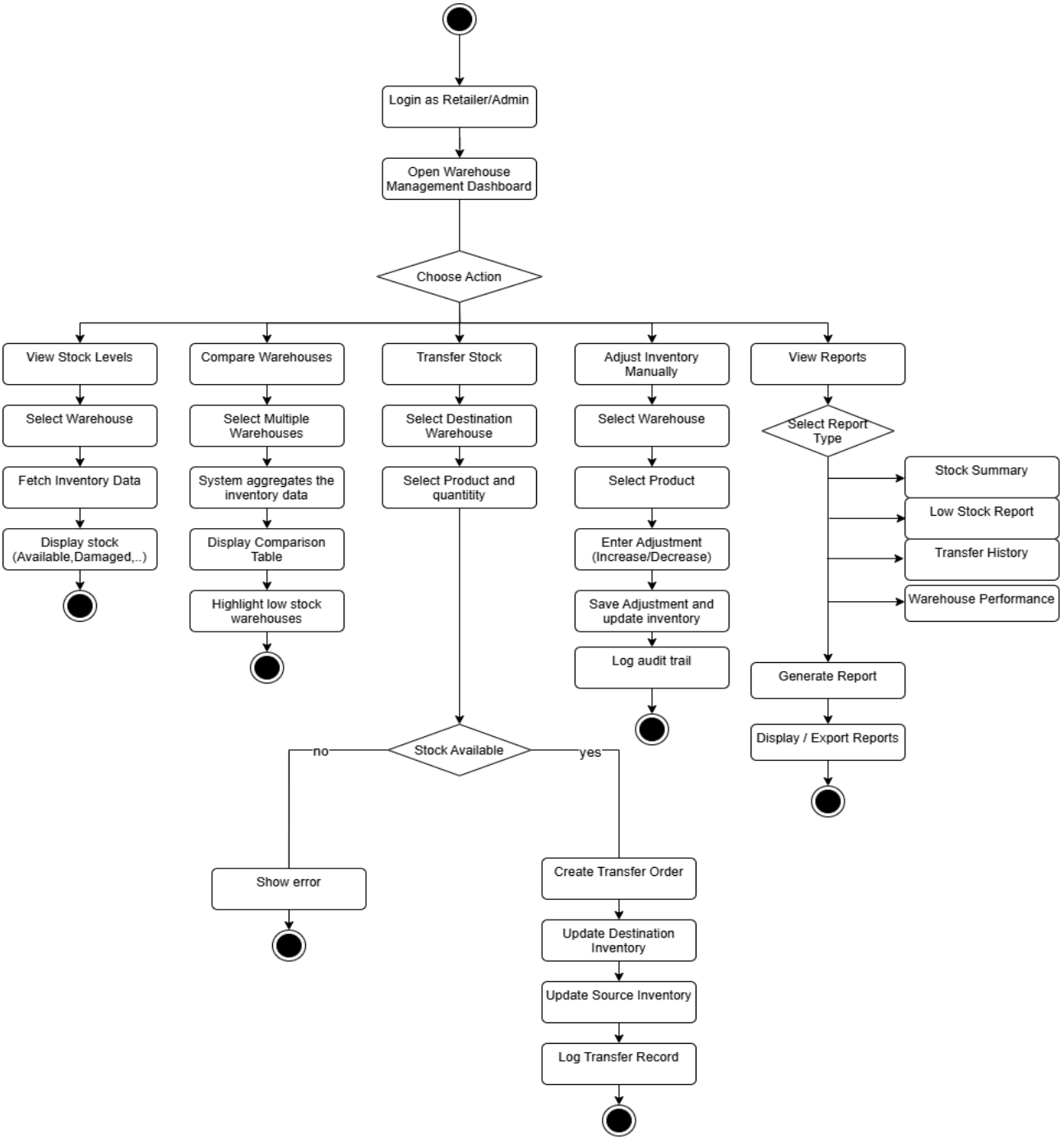
9.) Track Stock LIfecycle activity diagram:



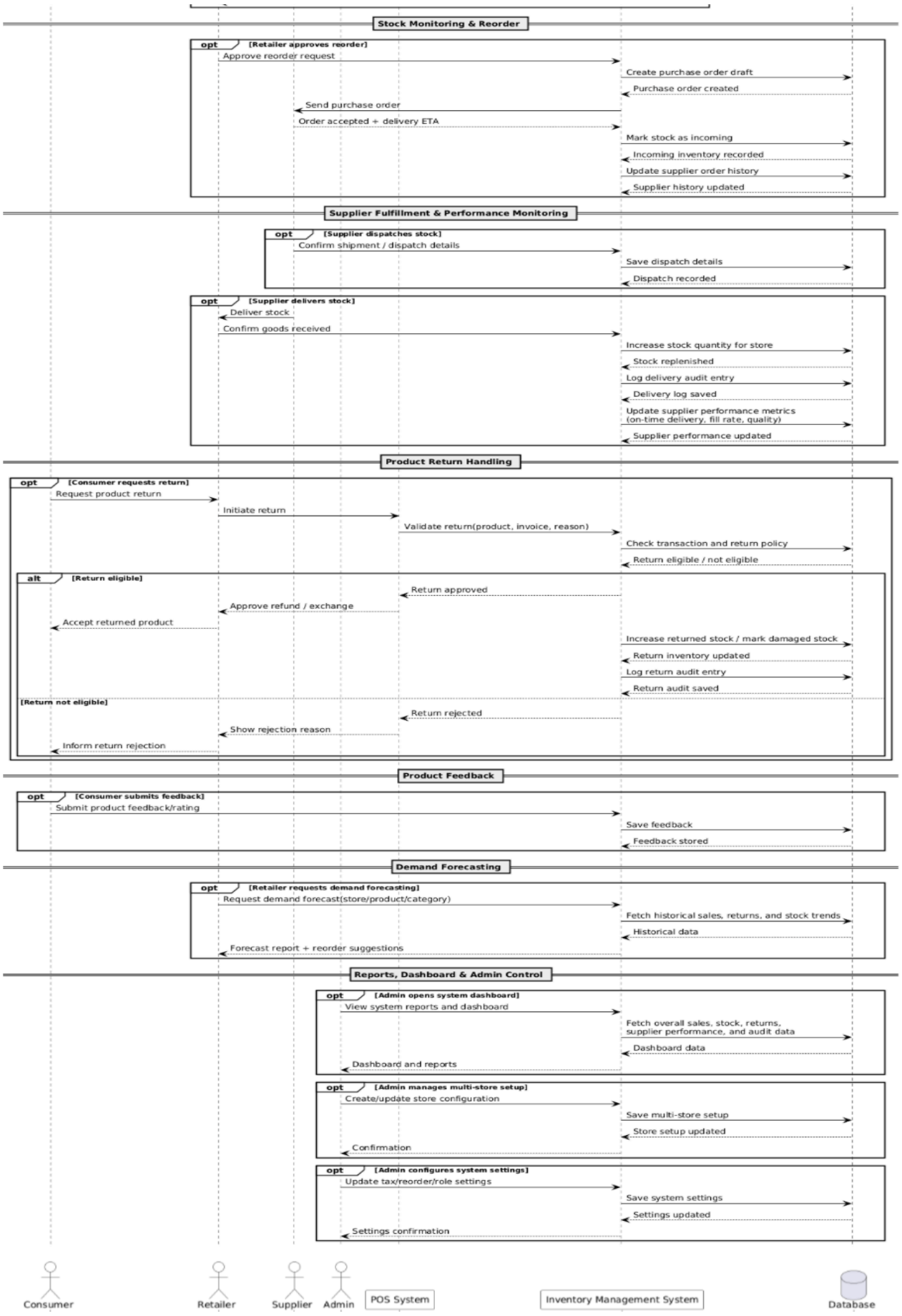
10.) Supplier Ordering and capacity management



11.) Multiple Warehouse management



Sequence Diagram:


Stock Monitoring & Reorder
opt [Retailer approves reorder]
Approve reorder request
Create purchase order draft
Purchase order created
Send purchase order
Order accepted + delivery ETA
Mark stock as incoming
Incoming inventory recorded
Update supplier order history
Supplier history updated
Supplier Fulfillment & Performance Monitoring
opt [Supplier dispatches stock]
Confirm shipment / dispatch details
Save dispatch details
Dispatch recorded
opt [Supplier delivers stock]
Deliver stock
Confirm goods received
Increase stock quantity for store
Stock replenished
Log delivery audit entry
Delivery log saved
Update supplier performance metrics
(on-time delivery, fill rate, quality)
Supplier performance updated
Product Return Handling
opt [Consumer requests return]
Request product return
Initiate return
Validate return(product, invoice, reason)
Check transaction and return policy
Return eligible / not eligible
alt [Return eligible]
Return approved
Approve refund / exchange
Accept returned product
Increase returned stock / mark damaged stock
Return inventory updated
Log return audit entry
Return audit saved
[Return not eligible]
Return rejected
Show rejection reason
Inform return rejection
Product Feedback
opt [Consumer submits feedback]
Submit product feedback/rating
Save feedback
Feedback stored
Demand Forecasting
opt [Retailer requests demand forecasting]
Request demand forecast(store/product/category)
Fetch historical sales, returns, and stock trends
Historical data
Forecast report + reorder suggestions
Reports, Dashboard & Admin Control
opt [Admin opens system dashboard]
View system reports and dashboard
Fetch overall sales, stock, returns,
supplier performance, and audit data
Dashboard data
Dashboard and reports
opt [Admin manages multi-store setup]
Create/update store configuration
Save multi-store setup
Store setup updated
Confirmation
opt [Admin configures system settings]
Update tax/reorder/role settings
Save system settings
Settings updated
Settings confirmation
Consumer
Retailer
Supplier
Admin
POS System
Inventory Management System
Database